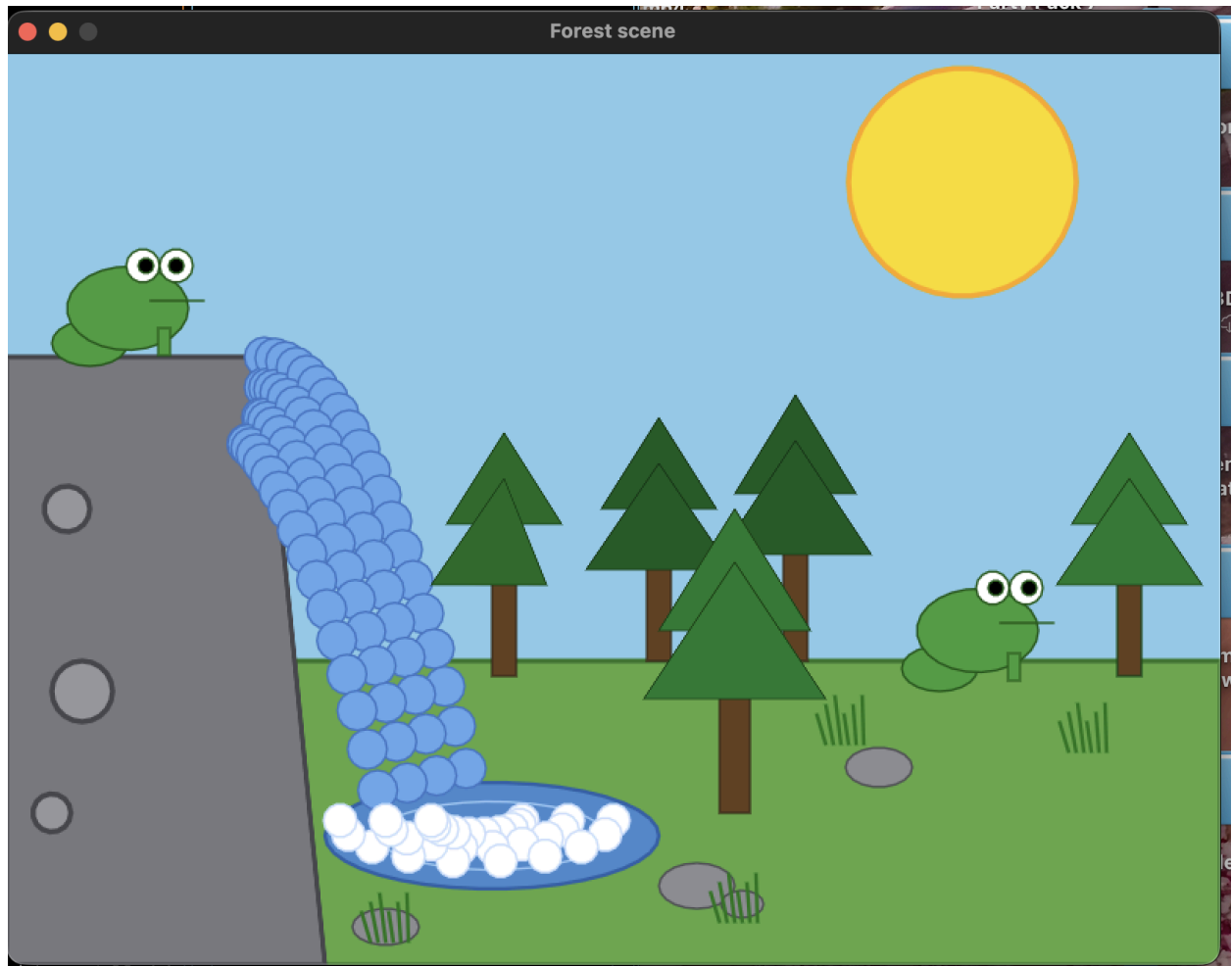
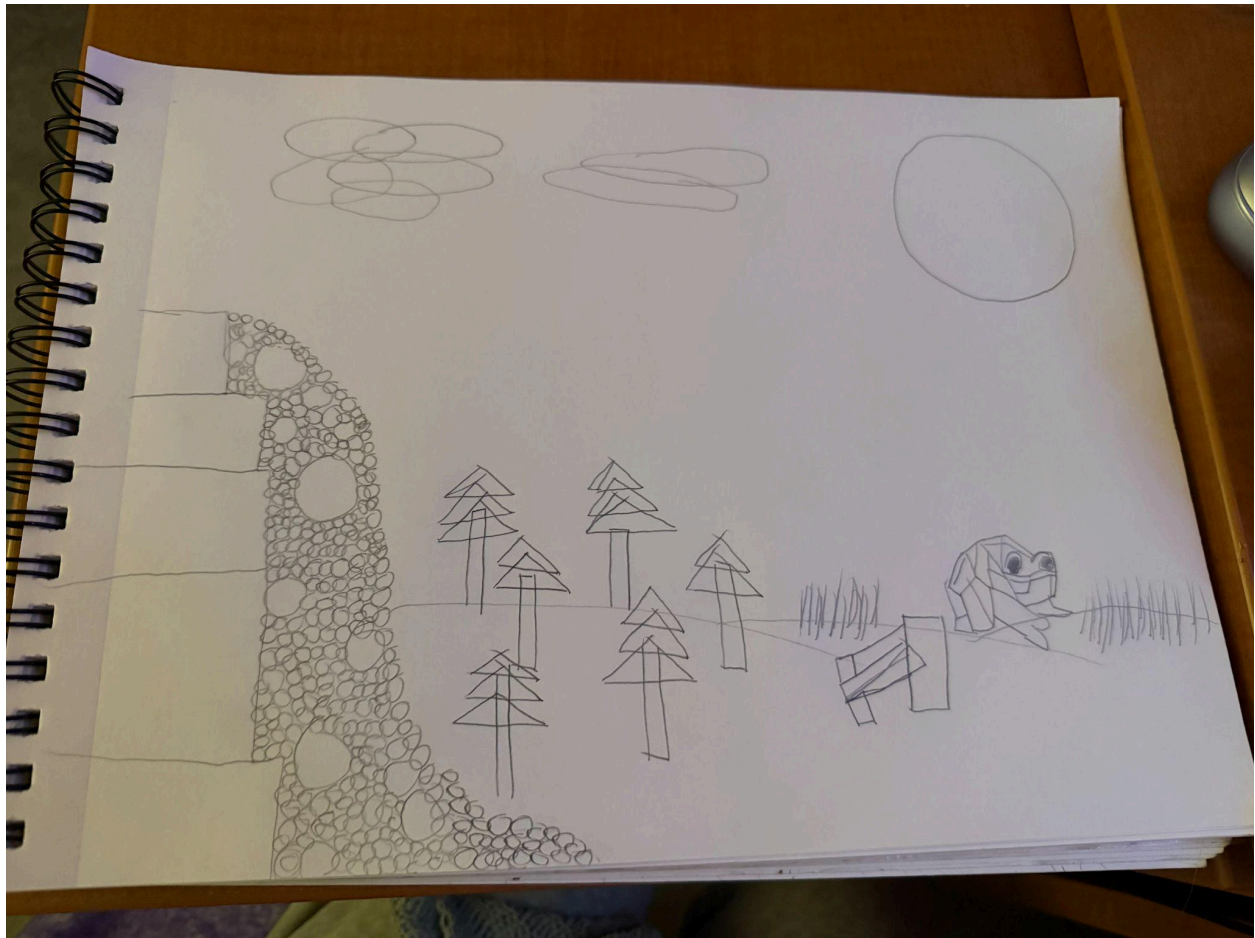






```
require("L5")

function setup()
  size(800, 600)

  -- Set the program title
  windowTitle("Forest scene")

  -- Set print command output to display in window
  printToScreen()

  describe( ' A calming waterfall with pine trees and a frog basking in the sun')
end

function draw()
  -- sky
  background(135, 206, 235)

  -- sun
  fill(255, 220, 0)
```

```
stroke(255, 170, 0)
strokeWeight(3)
circle(630, 85, 150)

-- ground
fill(90, 170, 70)
stroke(50, 120, 40)
rect(0, 400, 800, 200)

-- cliff
fill(120, 120, 125)
stroke(70, 70, 75)
quad(0, 200, 170, 200, 210, 600, 0, 600)

-- cliff rocks
fill(150, 150, 155)
circle(40, 300, 30)
circle(50, 420, 40)
circle(30, 500, 25)

-- pond at the base of the waterfall
fill(70, 140, 210)
stroke(40, 100, 170)
strokeWeight(2)
ellipse(320, 515, 220, 70)

-- pond ripples
noFill()
stroke(150, 200, 245)
strokeWeight(1)
ellipse(320, 515, 150, 45)
ellipse(320, 515, 80, 22)

-- waterfall stream
fill(100, 170, 240)
stroke(60, 120, 200)
strokeWeight(1)
for s = 0, 1, 0.05 do
  local x = 170 + s * 140
  local y = 200 + s * s * 300

  -- direction the water is moving at this point on the arc
```

```

local dx = 140
local dy = 600 * s
local len = sqrt(dx * dx + dy * dy)

-- four circles side by side across the width of the stream
for k = 0, 3 do
    circle(x - dy / len * k * 20, y + dx / len * k * 20, 26)
end
end

-- splash fanning out at the bottom
fill(255)
stroke(200, 225, 255)
for angle = 0, 180, 20 do
    for dist = 30, 90, 30 do
        local x = 310 + cos(radians(angle)) * dist
        local y = 505 + sin(radians(angle)) * dist * 0.3
        circle (x, y, 22)
    end
end
end

-- tree 1 trunk
fill (101, 67, 33)
stroke(60, 40, 20)
rect(320, 330, 16, 80)

-- tree 1 branches
fill(20, 110, 40)
stroke(10, 70, 20)
triangle(328, 250, 290, 310, 366, 310)
triangle(328, 280, 280, 350, 356, 350)

-- tree 2 trunk (back)
fill(101, 67, 33)
stroke(60, 40, 20)
strokeWeight(1)
rect(422, 320, 16, 80)

-- tree 2 branches
fill(15, 95, 35)
stroke(10, 60, 20)
triangle(430, 240, 392, 300, 468, 300)

```

```
triangle(430, 270, 382, 340, 478, 340)

-- tree 3 trunk (back)
fill(101, 67, 33)
stroke(60, 40, 20)
rect(512, 310, 16, 90)

-- tree 3 branches
fill(15, 95, 35)
stroke(10, 60, 20)
triangle(520, 225, 480, 290, 560, 290)
triangle(520, 255, 470, 330, 570, 330)

-- tree 4 trunk (front)
fill(101, 67, 33)
stroke(60, 40, 20)
rect(470, 400, 20, 100)

-- tree 4 branches
fill(25, 125, 45)
stroke(10, 70, 20)
triangle(480, 300, 430, 380, 530, 380)
triangle(480, 335, 420, 425, 540, 425)

-- tree 5 trunk (right edge)
fill(101, 67, 33)
stroke(10, 70, 20)
rect(732, 330, 16, 80)

-- tree 5 branches
fill(25, 125, 45)
stroke(10, 70, 20)
triangle(740, 250, 702, 310, 778, 310)
triangle(740, 280, 692, 350, 788, 350)

-- rocks
fill(140, 140, 145)
stroke(80, 80, 85)
ellipse(455, 548, 50, 30)
ellipse(485, 560, 28, 18)
ellipse(250, 575, 44, 24)
ellipse(575, 470, 44, 26)
```

```

-- grass tufts
stroke(30, 120, 30)
strokeWeight(2)
for i = 0, 5 do
  line(540 + i * 5, 455, 534 + i * 6, 435 - (i % 3) * 6)
  line(700 + i * 5, 460, 694 + i * 6, 440 - (i % 3) * 6)
  line(240 + i * 5, 585, 234 + i * 6, 565 - (i % 3) * 6)
  line(470 + i * 5, 572, 464 + i * 6, 552 - (i % 3) * 6)
end
strokeWeight(1)

-- frog back leg
fill(60, 160, 60)
stroke(20, 90, 20)
ellipse(615, 405, 50, 30)

-- frog body
ellipse(640, 380, 80, 55)

-- frog front leg
rect(660, 395, 8, 18)

-- frog eyes (white)
fill(255)
circle(650, 352, 22)
circle(672, 352, 22)

-- frog pupils
fill(0)
circle(652, 352, 10)
circle(674, 352, 10)

-- frog mouth
line(655, 375, 690, 375)

-- frog 2 back leg (On the cliff top)
fill(60, 160, 60)
stroke(20, 90, 20)
ellipse(55, 191, 50, 30)

-- frog 2 body

```

```
ellipse(80, 168, 80, 55)

-- frog 2 front leg
rect(100, 181, 8, 18)

-- frog 2 eyes (white)
fill(255)
circle(90, 140, 22)
circle(112, 140, 22)

-- frog 2 pupils
fill(0)
circle(92, 140, 10)
circle(112, 140, 10)

-- frog 2 mouth
line(95, 163, 130, 163)
end
```